# FlexGroup 卷的灾难恢复

ONTAP 9

NetApp
January 25, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/flexgroup/disaster-recovery-workflow-task.html on January 25, 2022. Always check docs.netapp.com for the latest.

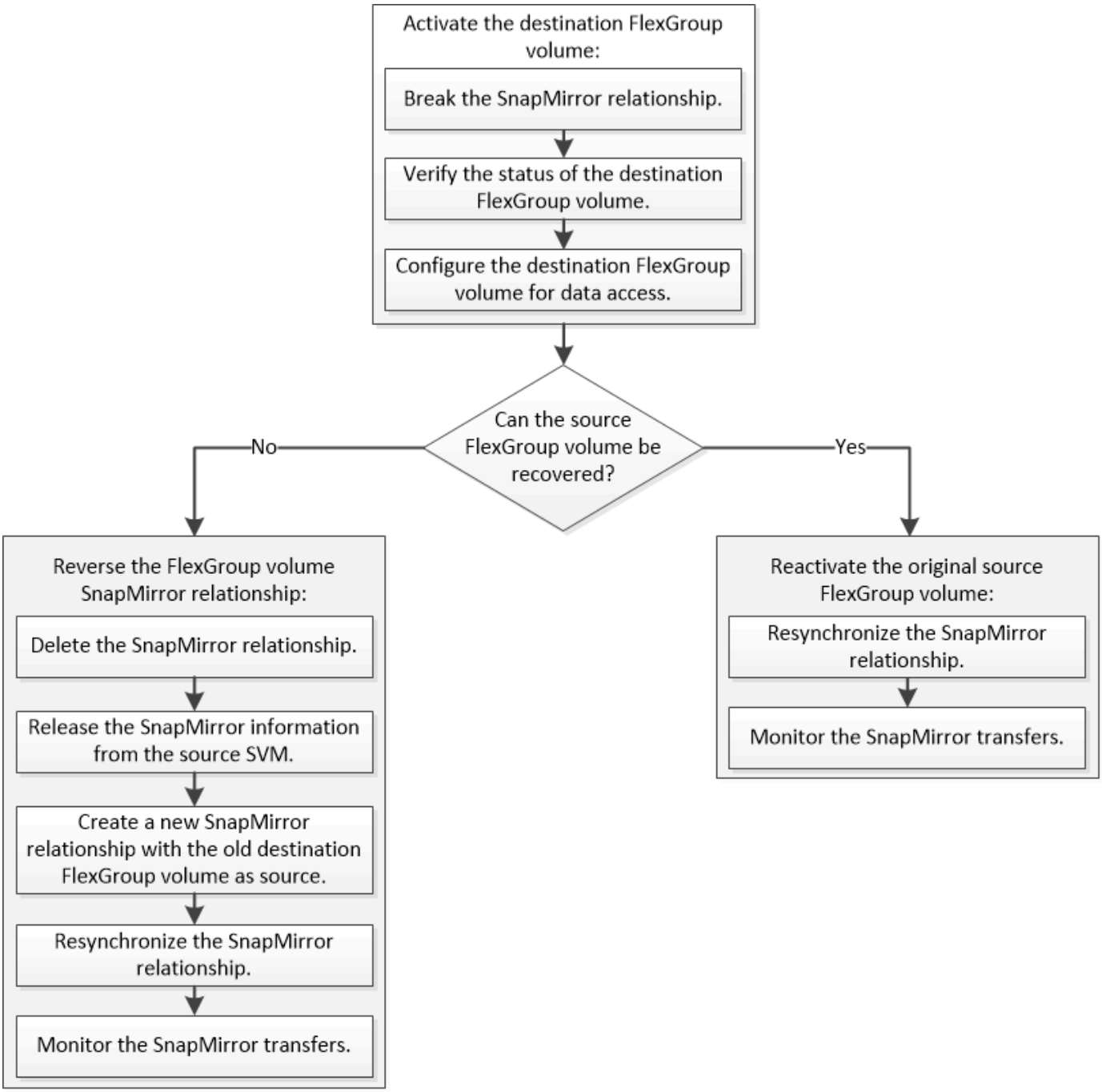
# 目录

FlexGroup 卷的灾难恢复 . . . . . 1
FlexGroup 卷的灾难恢复工作流 . . . . . 1
激活目标 FlexGroup 卷 . . . . . 2
发生灾难后重新激活原始源 FlexGroup 卷 . . . . . 4
在灾难恢复期间反转 FlexGroup 卷之间的 SnapMirror 关系 . . . . . 5

# FlexGroup 卷的灾难恢复

## FlexGroup 卷的灾难恢复工作流

当源 FlexGroup 卷发生灾难时，您应激活目标 FlexGroup 卷并重定向客户端访问。根据源 FlexGroup 卷是否可以恢复，您应重新激活源 FlexGroup 卷或反转 SnapMirror 关系。




当某些 SnapMirror 操作（例如 SnapMirror 中断和重新同步）正在运行时，客户端对目标 FlexGroup 卷的访问会暂时被阻止。如果 SnapMirror 操作失败，则某些成分卷可能仍处于此状态，并且对 FlexGroup 卷的访问被拒绝。在这种情况下，您必须重试 SnapMirror 操作。

# 激活目标 FlexGroup 卷

如果源 FlexGroup 卷由于数据损坏，意外删除或脱机状态等事件而无法提供数据，则必须激活目标 FlexGroup 卷以提供数据访问，直到恢复源 FlexGroup 卷上的数据为止。激活涉及停止未来的 SnapMirror 数据传输并中断 SnapMirror 关系。

您必须从目标集群执行此任务。

步骤

1. 禁用 FlexGroup 卷 SnapMirror 关系的未来传输： `snapmirror quiesce dest_sVM ： dest_flexgroup`

   ```
   cluster2::> snapmirror quiesce -destination-path vsd:dst
   ```

2. 断开 FlexGroup 卷 SnapMirror 关系： `snapmirror break dest_sVM ： dest_flexgroup`

   ```
   cluster2::> snapmirror break -destination-path vsd:dst
   ```

3. 查看 SnapMirror 关系的状态： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Broken-off
                                    Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Broken-off
                                    Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Broken-off
                                    Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Broken-off
                                    Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Broken-off
                                    Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Broken-off
                                    Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Broken-off
                                    Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Broken-off
                                    Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Broken-off
                                    Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态为 `broken-off` 。

4. 验证目标 FlexGroup 卷是否为读 / 写卷： `volume show -vserver svm_name`

```
cluster2::> volume show -vserver vsd
Vserver   Volume       Aggregate    State      Type       Size
Available Used%
--------- ------------ ------------ ---------- ---- ----------
---------- -----
vsd       dst          -            online     **RW**          2GB
1.54GB   22%
vsd       d2           -            online     DP          2GB
1.55GB   22%
vsd       root_vs0     aggr1        online     RW        100MB
94.02MB    5%
3 entries were displayed.
```

5. 将客户端重定向到目标 FlexGroup 卷。

# 发生灾难后重新激活原始源 FlexGroup 卷

当源 FlexGroup 卷变为可用时，您可以重新同步原始源 FlexGroup 卷和原始目标 卷。目标 FlexGroup 卷上的任何新数据将丢失。

在执行重新同步之前，目标卷上的任何活动配额规则都将被停用，并且配额规则将被删除。

重新同步操作完成后，您可以使用 `volume quota policy rule create` 和 `volume quota modify` 命令创建和重新激活配额规则。

步骤

1. 从目标集群重新同步 FlexGroup 卷 SnapMirror 关系： `snapmirror resync -destination-path dst_sVM ： dest_flexgroup`
2. 查看 SnapMirror 关系的状态： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Snapmirrored
                                      Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Snapmirrored
                                      Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Snapmirrored
                                      Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Snapmirrored
                                      Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Snapmirrored
                                      Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Snapmirrored
                                      Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Snapmirrored
                                      Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Snapmirrored
                                      Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Snapmirrored
                                      Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态为 `Snapmirrored` 。

# 在灾难恢复期间反转 FlexGroup 卷之间的 SnapMirror 关系

如果灾难禁用了 SnapMirror 关系中的源 FlexGroup 卷，则可以在修复或替换源 FlexGroup 卷时使用目标 FlexGroup 卷提供数据。源 FlexGroup 卷联机后，您可以将原始源 FlexGroup 卷设为只读目标，并反转 SnapMirror 关系。

在执行重新同步之前，目标卷上的任何活动配额规则都将被停用，并且配额规则将被删除。

重新同步操作完成后，您可以使用 `volume quota policy rule create` 和 `volume quota modify` 命令创建和重新激活配额规则。

步骤

1. 在初始目标 FlexGroup 卷上，删除源 FlexGroup 卷与目标 FlexGroup 卷之间的数据保护镜像关系：`snapmirror delete -destination-path svm_name： volume_name`

```
cluster2::> snapmirror delete -destination-path vsd:dst
```

2. 在原始源 FlexGroup 卷上，从源 FlexGroup 卷中删除关系信息： `snapmirror release -destination -path svm_name ： volume_name -relationship-info-only`

   删除 SnapMirror 关系后，必须先从源 FlexGroup 卷中删除关系信息，然后再尝试重新同步操作。

```
cluster1::> snapmirror release -destination-path vsd:dst -relationship
-info-only true
```

3. 在新的目标 FlexGroup 卷上，创建镜像关系： `snapmirror create -source-path src_svm_name ： volume_name -destination-path dst_svm_name ： volume_name -type XDP -policy MirrorAllSnapshots`

```
cluster1::> snapmirror create -source-path vsd:dst -destination-path
vss:src -type XDP -policy MirrorAllSnapshots
```

4. 在新的目标 FlexGroup 卷上，重新同步源 FlexGroup： `snapmirror resync -source-path svm_name ： volume_name`

```
cluster1::> snapmirror resync -source-path vsd:dst
```

5. 监控 SnapMirror 传输： `snapmirror show -expand`

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vsd:dst       XDP  vss:src        Snapmirrored
                                       Idle           -         true    -
vss:dst__0001 XDP  vss:src__0001  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0002 XDP  vss:src__0002  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0003 XDP  vss:src__0003  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0004 XDP  vss:src__0004  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0005 XDP  vss:src__0005  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0006 XDP  vss:src__0006  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0007 XDP  vss:src__0007  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0008 XDP  vss:src__0008  Snapmirrored
                                       Idle           -         true    -
...
```

每个成分卷的 SnapMirror 关系状态显示为 `Snapmirrored`，表示重新同步已成功。

**Copyright Information**

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

**Trademark Information**

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.